

# 个人游戏 Demo 性能优化记录

## 一、优化前的监控与分析

### 1. 首先查看总帧时间

在控制台输入 `stat unit` 和 `fps` 命令



#### 各项解读：

- 帧率 79，帧预算 12.84
- Game: 4.49，远小于 Frame → 逻辑侧无压力
- Draw: 12.83  $\approx$  Frame → 渲染线程耗时过高，瓶颈所在（Drawcall/状态切换）
- RHIT: 6.73 < Draw 但不算低 → CPU 到驱动的提交也有压力
- GPU: 6.07 < Frame $\times$ 85% → 达标，说明瓶颈不在 GPU 绘制

### 2. 深入分析渲染提交侧

从总体数据看，CPU 渲染提交侧是绝对瓶颈。RHI 层核心执行耗时，约占 Draw 的一半，说明渲染线程的图元收集/命令生成（非 RHI）也占了约 6ms  
查看 RHI 层计数器数值。

| RHI [STA] GROUP [TH]                     |  | CallCount  | InclusiveAvg | InclusiveMax | ExclusiveAvg  | ExclusiveMax |
|------------------------------------------|--|------------|--------------|--------------|---------------|--------------|
| Cycle counters (flat)                    |  |            |              |              |               |              |
| FRDGPass Execute                         |  | 680        | 4.81 ms      | 5.04 ms      | 3.80 ms       | 4.44 ms      |
| Memory Counters                          |  | UsedMax    | Mem%         | MemPool      | Pool Capacity |              |
| Reserved Buffer Memory (Uncommitted)     |  | 1520.00 MB |              | GPU          |               |              |
| Reserved Buffer Memory (Committed)       |  | 528.00 MB  |              | GPU          |               |              |
| UAV Texture Memory                       |  | 415.38 MB  |              | GPU          |               |              |
| Texture 2D Memory                        |  | 348.26 MB  |              | GPU          |               |              |
| Render Target 2D Memory                  |  | 349.69 MB  |              | GPU          |               |              |
| Reserved Texture Memory (Committed)      |  | 256.00 MB  |              | GPU          |               |              |
| RayTracing Acceleration Structure Memory |  | 66.44 MB   |              | GPU          |               |              |
| Vertex Buffer Memory                     |  | 64.50 MB   |              | GPU          |               |              |
| Bindless Resource Heap                   |  | 37.71 MB   |              | GPU          |               |              |
| Structured Buffer Memory                 |  | 24.92 MB   |              | GPU          |               |              |
| Texture Cube Memory                      |  | 21.31 MB   |              | GPU          |               |              |
| Index Buffer Memory                      |  | 10.14 MB   |              | GPU          |               |              |
| Byte Address Buffer Memory               |  | 4.71 MB    |              | GPU          |               |              |
| Texture 3D Memory                        |  | 2.36 MB    |              | GPU          |               |              |
| Bindless Sampler Heap                    |  | 1.57 MB    |              | GPU          |               |              |
| Uniform Buffer Memory                    |  | 0.79 MB    |              | GPU          |               |              |
| Render Target 3D Memory                  |  | 0.19 MB    |              | GPU          |               |              |
| Misc Buffer Memory                       |  | 0.00 MB    |              | GPU          |               |              |
| Draw Indirect Buffer Memory              |  | 0.00 MB    |              | GPU          |               |              |
| Render Target Cube Memory                |  | 0.00 MB    |              | GPU          |               |              |
| Reserved Texture Memory (Uncommitted)    |  | 0.00 MB    |              | GPU          |               |              |
| Counters                                 |  | Average    | Max          | Min          |               |              |
| Resource Descriptors Allocated           |  |            | 505,449.00   | 505,429.00   |               |              |
| Triangles drawn                          |  | 101,283.20 | 230,404.00   | 87,520.00    |               |              |
| Bindless Resource Descriptors Allocated  |  |            | 5,449.00     | 5,429.00     |               |              |
| Sampler Descriptors Allocated            |  |            | 2,092.00     | 2,092.00     |               |              |
| DrawPrimitive calls                      |  | 537.70     | 556.00       | 533.00       |               |              |
| Bindless Sampler Descriptors Allocated   |  |            | 44.00        | 44.00        |               |              |
| Lines drawn                              |  | 4.00       | 4.00         | 4.00         |               |              |

## Counters

### 1. Triangles drawn

平均 101K，与 Prims: 192K 同量级，合理，说明不是单纯三角数压爆，**提交/状态/资源绑定/可见性收集/可能材质或绑定开销**更值得关注。

### 2. DrawPrimitive calls

平均 537，与 Draws: 436 差距较小，在 PC 端不一定不行，但结合 Render/Draw 12.83ms，说明还有优化空间。

平均每个 Drawcall 渲染  $\frac{101K}{537} \approx 188$  个三角面，属于典型“小批次”问题。接近 500 次绘制中，大量开销消耗在材质切换、动态图元收集与渲染状态设置上，而非实际光栅化（GPU 仅 6.07ms，非常轻松）

### 3. Resource Descriptors Allocated

506K，这是 RHI/GPU 资源描述符分配计数，数值很高。Bindless 开/平台 D3D12/Vulkan 下会更常见，但它是信号：资源绑定复杂度不低。

## Memory Counters

### 1. Texture 2D: 348MB

贴图占用内存合理。

### 2. UAV Texture: 415MB

合理，考虑到场景实际情况，最可能与体积雾/后处理有关。

### 3. Render Target 2D: 349MB

在 Standalone 模式下测试，观察 memreport 得到以下数据

| RHI resource memory (not tracked by our allocator) |                                                                                                                    |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 41.768MB                                           | - Bindless Resource Heap - STAT_BindlessResourceHeapMemory - STATGROUP_RHI - STATCAT_Advanced                      |
| 1.566MB                                            | - Bindless Sampler Heap - STAT_BindlessSamplerHeapMemory - STATGROUP_RHI - STATCAT_Advanced                        |
| 256.000MB                                          | - Reserved Texture Memory (Committed) - STAT_ReservedCommittedTextureMemory - STATGROUP_RHI - STATCAT_Advanced     |
| 0.000MB                                            | - Reserved Texture Memory (Uncommitted) - STAT_ReservedUncommittedTextureMemory - STATGROUP_RHI - STATCAT_Advanced |
| 0.000MB                                            | - Reserved Buffer Memory (Committed) - STAT_ReservedCommittedBufferMemory - STATGROUP_RHI - STATCAT_Advanced       |
| 2048.000MB                                         | - Reserved Buffer Memory (Uncommitted) - STAT_ReservedUncommittedBufferMemory - STATGROUP_RHI - STATCAT_Advanced   |
| 0.000MB                                            | - Misc Buffer Memory - STAT_MiscBufferMemory - STATGROUP_RHI - STATCAT_Advanced                                    |
| 0.000MB                                            | - Draw Indirect Buffer Memory - STAT_DrawIndirectBufferMemory - STATGROUP_RHI - STATCAT_Advanced                   |
| 4.713MB                                            | - Byte Address Buffer Memory - STAT_ByteAddressBufferMemory - STATGROUP_RHI - STATCAT_Advanced                     |
| 24.641MB                                           | - Structured Buffer Memory - STAT_StructuredBufferMemory - STATGROUP_RHI - STATCAT_Advanced                        |
| 66.075MB                                           | - RayTracing Acceleration Structure Memory - STAT_RTAccelerationStructureMemory - STATGROUP_RHI - STATCAT_Advanced |
| 63.695MB                                           | - Vertex Buffer Memory - STAT_VertexBufferMemory - STATGROUP_RHI - STATCAT_Advanced                                |
| 10.102MB                                           | - Index Buffer Memory - STAT_IndexBufferMemory - STATGROUP_RHI - STATCAT_Advanced                                  |
| 0.786MB                                            | - Uniform Buffer Memory - STAT_UniformBufferMemory - STATGROUP_RHI - STATCAT_Advanced                              |
| 21.309MB                                           | - Texture Cube Memory - STAT_TextureMemoryCube - STATGROUP_RHI - STATCAT_Advanced                                  |
| 2.359MB                                            | - Texture 3D Memory - STAT_TextureMemory3D - STATGROUP_RHI - STATCAT_Advanced                                      |
| 346.738MB                                          | - Texture 2D Memory - STAT_TextureMemory2D - STATGROUP_RHI - STATCAT_Advanced                                      |
| 343.875MB                                          | - UAV Texture Memory - STAT_UAVTextureMemory - STATGROUP_RHI - STATCAT_Advanced                                    |
| 0.000MB                                            | - Render Target Cube Memory - STAT_RenderTargetMemoryCube - STATGROUP_RHI - STATCAT_Advanced                       |
| 0.188MB                                            | - Render Target 3D Memory - STAT_RenderTargetMemory3D - STATGROUP_RHI - STATCAT_Advanced                           |
| 342.188MB                                          | - Render Target 2D Memory - STAT_RenderTargetMemory2D - STATGROUP_RHI - STATCAT_Advanced                           |
| 3574.003MB total                                   |                                                                                                                    |
| MemReport: End command "rhi.DumpMemory"            |                                                                                                                    |

RHI resource memory: 约 3.6GB 远小于显存大小，无风险

|                                                                    |
|--------------------------------------------------------------------|
| > Pooled Render Targets: ...                                       |
| 865.980MB total, 810.480MB used, 0.000MB unused, 79 render targets |

Pooled Render Targets: 865MB，远小于显存，无风险

结合 Stat Unit 的 GPU 时间来看，总体而言，无显存泄露风险。

### 3. 拆解渲染线程

➤ 查看 Stat SceneRendering 数据

| Scene Rendering [STATGROUP_SceneRendering]                           |  | CallCount | InclusiveAvg | InclusiveMax | ExclusiveAvg | ExclusiveMax |
|----------------------------------------------------------------------|--|-----------|--------------|--------------|--------------|--------------|
| Cycle counters (flat)                                                |  |           |              |              |              |              |
| RenderViewFamily                                                     |  | 1         | 10.64ms      | 14.93ms      | 3.42ms       | 4.04ms       |
| InitViews                                                            |  | 1         | 1.94ms       | 5.27ms       | 0.10ms       | 0.29ms       |
| DeferredShadingSceneRenderer_Lighting                                |  | 1         | 1.76ms       | 2.01ms       | 0.24ms       | 0.38ms       |
| Lighting drawing                                                     |  | 1         | 1.41ms       | 1.62ms       | 0.00ms       | 0.00ms       |
| Finish Gather Ray Tracing Instances                                  |  | 1         | 0.79ms       | 1.52ms       | 0.32ms       | 0.59ms       |
| Proj Shadow drawing                                                  |  | 74        | 0.14ms       | 0.21ms       | 0.05ms       | 0.07ms       |
| Dynamic shadow setup                                                 |  | 2         | 0.12ms       | 0.21ms       | 0.11ms       | 0.20ms       |
| Base pass drawing                                                    |  | 1         | 0.10ms       | 0.12ms       | 0.10ms       | 0.12ms       |
| DeferredShadingSceneRenderer_PrepForwardLightData                    |  | 1         | 0.09ms       | 0.16ms       | 0.01ms       | 0.02ms       |
| Translucency drawing                                                 |  | 1         | 0.04ms       | 0.05ms       | 0.04ms       | 0.05ms       |
| BeginOcclusionTests                                                  |  | 1         | 0.04ms       | 0.08ms       | 0.04ms       | 0.08ms       |
| DeferredShadingSceneRenderer_ViewExtensionPreRenderView              |  | 1         | 0.04ms       | 0.13ms       | 0.04ms       | 0.13ms       |
| DeferredShadingSceneRenderer_RenderFinish                            |  | 1         | 0.03ms       | 0.04ms       | 0.02ms       | 0.03ms       |
| DeferredShadingSceneRenderer_FXSystemPreRender                       |  | 1         | 0.03ms       | 0.07ms       | 0.03ms       | 0.07ms       |
| Depth drawing                                                        |  | 1         | 0.02ms       | 0.04ms       | 0.02ms       | 0.04ms       |
| DeferredShadingSceneRenderer_RenderSkyAtmosphere                     |  | 1         | 0.02ms       | 0.02ms       | 0.02ms       | 0.02ms       |
| InitViewsPossiblyAfterPrepass                                        |  | 1         | 0.02ms       | 0.03ms       | 0.02ms       | 0.03ms       |
| DeferredShadingSceneRenderer_ViewExtensionPostRenderView             |  | 1         | 0.01ms       | 0.02ms       | 0.01ms       | 0.02ms       |
| DeferredShadingSceneRenderer_OBuffer                                 |  | 1         | 0.01ms       | 0.02ms       | 0.01ms       | 0.02ms       |
| DeferredShadingSceneRenderer_RenderInit                              |  | 1         | 0.01ms       | 0.03ms       | 0.01ms       | 0.03ms       |
| DeferredShadingSceneRenderer_RenderFog                               |  | 1         | 0.01ms       | 0.01ms       | 0.01ms       | 0.01ms       |
| DeferredShadingSceneRenderer_AfterBasePass                           |  | 1         | 0.01ms       | 0.01ms       | 0.01ms       | 0.01ms       |
| OcclusionSubmittedFenceWait                                          |  | 1         | 0.01ms       | 0.20ms       | 0.00ms       | 0.00ms       |
| RenderVelocities                                                     |  | 2         | 0.00ms       | 0.01ms       | 0.00ms       | 0.01ms       |
| Wait Ray Tracing Dynamic Bindings                                    |  | 1         | 0.00ms       | 0.01ms       | 0.00ms       | 0.01ms       |
| [7 more stats. Use the stats.MaxPerGroup CVar to increase the limit] |  |           |              |              |              |              |
| Counters                                                             |  | Average   | Max          | Min          |              |              |
| Ray tracing dynamic update primitives                                |  | 32,768.00 | 32,768.00    | 32,768.00    |              |              |
| Ray tracing dynamic build primitives                                 |  | 14,328.00 | 15,144.00    | 12,984.00    |              |              |
| Ray tracing active instances                                         |  | 1,913.00  | 1,913.00     | 1,913.00     |              |              |
| Ray tracing total instances                                          |  | 1,913.00  | 1,913.00     | 1,913.00     |              |              |
| Mesh draw calls                                                      |  | 232.65    | 250.00       | 226.00       |              |              |
| Present time                                                         |  | 0.02ms    | 0.03ms       |              |              |              |
| Lights in scene                                                      |  |           |              | 88.00        |              |              |
| Decals in scene                                                      |  |           | 0.00         | 0.00         |              |              |

从图中顶部数据看，RenderViewFamily 是所有渲染 Pass 的容器，其内部主要子项应包括：

| 子项                  | 耗时占比  | 说明                |
|---------------------|-------|-------------------|
| InitViews           | 最大头   | 可见性计算、动态图元收集、遮挡查询 |
| PrePass / DepthPass | 中等    | 深度预渲染（受 Draws 影响） |
| BasePass            | 中等    | 不透明物体绘制命令生成       |
| Lighting (Deferred) | 依赖灯光数 | 延迟光照 Pass         |
| ShadowDepth         | 可能很高  | 动态阴影（需重点排查）       |
| Translucency        | 半透明物体 | 粒子/玻璃/雾           |
| PostProcessing      | 较低    | 后处理链              |

关键判断：若 InitViews + ShadowDepth 占大部分，则属于“可见性与阴影驱动的高 Draw 成本”；若 BasePass 主导，则是"材质/合批问题"。

➤ 进一步查看 Stat InitViews

| Init Views [STATGROUP_InitViews] |           |              |              |              |              |
|----------------------------------|-----------|--------------|--------------|--------------|--------------|
| Cycle counters (flat)            |           |              |              |              |              |
|                                  | CallCount | InclusiveAvg | InclusiveMax | ExclusiveAvg | ExclusiveMax |
| Compute View Relevance           | 53        | 1.60 ms      | 2.38 ms      | 1.60 ms      | 2.38 ms      |
| View Visibility                  | 1         | 1.41 ms      | 1.95 ms      | 0.15 ms      | 0.28 ms      |
| Init dynamic shadows             | 2         | 1.15 ms      | 1.77 ms      | 0.05 ms      | 0.07 ms      |
| Create WholeScene Shadow         | 77        | 0.86 ms      | 1.41 ms      | 0.86 ms      | 1.41 ms      |
| Add View Whole Scene Shadows     | 77        | 0.06 ms      | 0.08 ms      | 0.06 ms      | 0.08 ms      |
| Init Projected Shadow            | 1         | 0.01 ms      | 0.03 ms      | 0.01 ms      | 0.03 ms      |
| GatherShadowPrimitives           | 1         | 0.01 ms      | 0.01 ms      | 0.01 ms      | 0.01 ms      |
| Update Fading                    | 3         | 0.00 ms      | 0.01 ms      | 0.00 ms      | 0.01 ms      |
| Update Preshadow Cache           | 1         | 0.00 ms      | 0.00 ms      | 0.00 ms      | 0.00 ms      |
| Decompress Occlusion             | 1         | 0.00 ms      | 0.00 ms      | 0.00 ms      | 0.00 ms      |
| Counters                         |           |              |              |              |              |
|                                  | Average   | Max          | Min          |              |              |
| Processed primitives             | 2,485.00  | 2,485.00     | 2,485.00     |              |              |
| Visible Raytracing Primitives    | 1,912.00  | 1,912.00     | 1,912.00     |              |              |
| Frustum Culled primitives        | 613.00    | 613.00       | 613.00       |              |              |
| Occluded primitives              | 197.00    | 197.00       | 197.00       |              |              |
| Occlusion queries                | 49.00     | 49.00        | 49.00        |              |              |
| Visible static mesh elements     | 6.00      | 6.00         | 6.00         |              |              |
| Visible dynamic primitives       | 4.00      | 4.00         | 4.00         |              |              |

关键发现：

- 阴影总开销（Init dynamic + Create/Add WholeScene）≈ 2.07ms，占 InitViews 40%
- 可见性相关（Compute Relevance + View Visibility）≈ 3.01ms，占 InitViews 58%

图元（Primitives）数据揭示的深层问题

| 计数器                           | 数值   | 解读                      |
|-------------------------------|------|-------------------------|
| Processed primitives          | 2485 | 每帧处理总量大                 |
| Visible Raytracing Primitives | 1912 | 光追可见图元极高(场景可能开启了光追相关设置) |
| Visible static mesh elements  | 6    | 静态网格元素极少(合批严重不足，静态物未合并) |

## 4. 瓶颈定位与优化方向

### ● 核心矛盾：

RenderViewFamily 10.64ms 里，每一毫秒都在 CPU 端准备绘制命令，而 GPU 只用了 5.89ms——典型的"CPU 提交跟不上 GPU 渲染"。

- 优先级优化项:

1. 第一优先: **InitViews** (可见性收集)

- 症状: 每次 Tick 重新收集所有动态物体的可见性, 灯光/阴影交互多
- 优化:
  - 减少 `bTickEnabled` 的 Actor 数量
  - 静态物体标记 `Static` 并启用 `bUseAsOccluder`
  - 设置合理的 `Cull Distance` (距离剔除), 尤其植被/小物件
  - 合并小物体为 `ISM/HISM` (减少图元数量)

2. 第二优先: **ShadowDepth** (动态阴影)

- 症状: 动态光 + 大量投射阴影的物体 → 阴影 `Pass` 翻倍 `Draw Call`
- 优化:
  - 关闭非必要物体的 `Cast Shadow`
  - 减少动态光源数量, 改用静态光照/光照贴图
  - 降低阴影距离 (`r.Shadow.DistanceScale`)
  - 级联阴影 (`CSM`) 减少级联数

3. 第三优先: **BasePass** (绘制命令生成)

- 症状: 近 500 `Draws`, 平均每个仅 ~188 面, 批次过碎
- 优化:
  - 合并材质 (减少 `Material Slot`)
  - 静态网格合批 (`Merge Actors / HLOD`)
  - 启用 `Nanite` (静态高精度物体)
  - 共享 `Material Instance`, 减少状态切换

4. 第四优先: **Lighting / Translucency**

- 场景灯光多, 考虑静态化或减少动态灯
- 半透明物体 (粒子/玻璃) 走单独渲染路径, 避免打断合批

## 二、具体优化项前后对比

### 1. 临时关闭阴影

|                                          |            |              |              |               |              |                             |
|------------------------------------------|------------|--------------|--------------|---------------|--------------|-----------------------------|
| RHI [STATGROUP_rhi]                      |            |              |              |               |              | 79.18 FPS                   |
| Cycle counters (flat)                    |            |              |              |               |              | 12.63 ms                    |
| FRDGPass_Execute                         | CallCount  | InclusiveAvg | InclusiveMax | ExclusiveAvg  | ExclusiveMax | 12.68 ms                    |
|                                          | 684        | 4.10 ms      | 4.95 ms      | 3.63 ms       | 4.91 ms      | Frame: 12.67 ms             |
| Memory Counters                          |            |              |              |               |              | Game: 5.82 ms               |
|                                          | UsedMax    | Mem%         | MemPool      | Pool Capacity |              | Draw: 12.67 ms              |
| Reserved Buffer Memory (Uncommitted)     | 1520.00 MB |              | GPU          |               |              | RHIT: 6.40 ms               |
| Reserved Buffer Memory (Committed)       | 528.00 MB  |              | GPU          |               |              | GPU Time: 6.43 ms           |
| Texture 2D Memory                        | 346.74 MB  |              | GPU          |               |              | Input: 39.65 ms             |
| UAV Texture Memory                       | 343.88 MB  |              | GPU          |               |              | Mem: 4.12 GB                |
| Render Target 2D Memory                  | 342.10 MB  |              | GPU          |               |              | RenderRes: 87.0% (1114x627) |
| Reserved Texture Memory (Committed)      | 256.00 MB  |              | GPU          |               |              | Draws: 497                  |
| RayTracing Acceleration Structure Memory | 66.56 MB   |              | GPU          |               |              | Prims: 89.4K                |
| Vertex Buffer Memory                     | 64.60 MB   |              | GPU          |               |              |                             |
| Bindless Resource Heap                   | 37.71 MB   |              | GPU          |               |              |                             |
| Structured Buffer Memory                 | 24.61 MB   |              | GPU          |               |              |                             |
| Texture Cube Memory                      | 21.31 MB   |              | GPU          |               |              |                             |
| Index Buffer Memory                      | 10.14 MB   |              | GPU          |               |              |                             |
| Byte Address Buffer Memory               | 4.72 MB    |              | GPU          |               |              |                             |
| Texture 3D Memory                        | 2.36 MB    |              | GPU          |               |              |                             |
| Bindless Sampler Heap                    | 1.57 MB    |              | GPU          |               |              |                             |
| Uniform Buffer Memory                    | 1.28 MB    |              | GPU          |               |              |                             |
| Render Target 3D Memory                  | 0.19 MB    |              | GPU          |               |              |                             |
| Misc Buffer Memory                       | 0.00 MB    |              | GPU          |               |              |                             |
| Draw Indirect Buffer Memory              | 0.00 MB    |              | GPU          |               |              |                             |
| Render Target Cube Memory                | 0.00 MB    |              | GPU          |               |              |                             |
| Reserved Texture Memory (Uncommitted)    | 0.00 MB    |              | GPU          |               |              |                             |
| Counters                                 |            |              |              |               |              |                             |
|                                          | Average    | Max          | Min          |               |              |                             |
| Resource Descriptors Allocated           |            | 505,583.00   | 505,519.00   |               |              |                             |
| Triangles drawn                          | 102,799.34 | 232,234.00   | 89,356.00    |               |              |                             |
| Bindless Resource Descriptors Allocated  |            | 5,583.00     | 5,619.00     |               |              |                             |
| Sampler Descriptors Allocated            |            | 2,092.00     | 2,092.00     |               |              |                             |
| DrawPrimitive calls                      | 502.30     | 512.00       | 497.00       |               |              |                             |
| Bindless Sampler Descriptors Allocated   |            | 44.00        | 44.00        |               |              |                             |
| Lines drawn                              | 4.00       | 4.00         | 4.00         |               |              |                             |

|                                                                      |           |              |              |              |              |
|----------------------------------------------------------------------|-----------|--------------|--------------|--------------|--------------|
| Scene Rendering [STATGROUP_SceneRendering]                           |           |              |              |              |              |
| Cycle counters (flat)                                                |           |              |              |              |              |
|                                                                      | CallCount | InclusiveAvg | InclusiveMax | ExclusiveAvg | ExclusiveMax |
| RenderViewFamily                                                     | 1         | 9.97 ms      | 14.38 ms     | 3.41 ms      | 5.84 ms      |
| DeferredShadingSceneRenderer Lighting                                | 1         | 1.78 ms      | 2.36 ms      | 0.24 ms      | 0.50 ms      |
| Lighting drawing                                                     | 1         | 1.44 ms      | 1.91 ms      | 0.00 ms      | 0.00 ms      |
| InitViews                                                            | 1         | 1.28 ms      | 2.60 ms      | 0.09 ms      | 0.26 ms      |
| Finish Gather Ray Tracing Instances                                  | 1         | 0.87 ms      | 1.18 ms      | 0.32 ms      | 0.48 ms      |
| Proj Shadow drawing                                                  | 70        | 0.14 ms      | 0.21 ms      | 0.05 ms      | 0.07 ms      |
| Dynamic shadow setup                                                 | 2         | 0.11 ms      | 0.16 ms      | 0.10 ms      | 0.15 ms      |
| Base pass drawing                                                    | 1         | 0.10 ms      | 0.16 ms      | 0.10 ms      | 0.16 ms      |
| DeferredShadingSceneRenderer PrepareForwardLightData                 | 1         | 0.09 ms      | 0.14 ms      | 0.01 ms      | 0.02 ms      |
| Translucency drawing                                                 | 1         | 0.04 ms      | 0.05 ms      | 0.04 ms      | 0.05 ms      |
| BeginOcclusionTests                                                  | 1         | 0.04 ms      | 0.10 ms      | 0.04 ms      | 0.10 ms      |
| DeferredShadingSceneRenderer FXSystem PreRender                      | 1         | 0.03 ms      | 0.06 ms      | 0.03 ms      | 0.06 ms      |
| DeferredShadingSceneRenderer ViewExtensionPreRenderView              | 1         | 0.04 ms      | 0.08 ms      | 0.04 ms      | 0.08 ms      |
| DeferredShadingSceneRenderer RenderFinish                            | 1         | 0.03 ms      | 0.05 ms      | 0.02 ms      | 0.03 ms      |
| Depth drawing                                                        | 1         | 0.02 ms      | 0.04 ms      | 0.02 ms      | 0.04 ms      |
| DeferredShadingSceneRenderer RenderSkyAtmosphere                     | 1         | 0.02 ms      | 0.02 ms      | 0.02 ms      | 0.02 ms      |
| InitViewsPossiblyAfterPrepass                                        | 1         | 0.02 ms      | 0.03 ms      | 0.02 ms      | 0.03 ms      |
| DeferredShadingSceneRenderer ViewExtensionPostRenderView             | 1         | 0.01 ms      | 0.02 ms      | 0.01 ms      | 0.02 ms      |
| DeferredShadingSceneRenderer DBuffer                                 | 1         | 0.01 ms      | 0.03 ms      | 0.01 ms      | 0.03 ms      |
| DeferredShadingSceneRenderer Render Init                             | 1         | 0.01 ms      | 0.05 ms      | 0.01 ms      | 0.05 ms      |
| DeferredShadingSceneRenderer RenderFog                               | 1         | 0.01 ms      | 0.01 ms      | 0.01 ms      | 0.01 ms      |
| DeferredShadingSceneRenderer AfterBasePass                           | 1         | 0.01 ms      | 0.01 ms      | 0.01 ms      | 0.01 ms      |
| RenderVelocities                                                     | 2         | 0.00 ms      | 0.01 ms      | 0.00 ms      | 0.01 ms      |
| Wait Ray Tracing Dynamic Bindings                                    | 1         | 0.00 ms      | 0.01 ms      | 0.00 ms      | 0.01 ms      |
| DeferredShadingSceneRenderer RenderLightShaftBloom                   | 1         | 0.00 ms      | 0.00 ms      | 0.00 ms      | 0.00 ms      |
| [7 more stats. Use the stats.MaxPerGroup CVar to increase the limit] |           |              |              |              |              |
| Counters                                                             |           |              |              |              |              |
|                                                                      | Average   | Max          | Min          |              |              |
| Ray tracing dynamic update primitives                                | 32,769.00 | 32,769.00    | 32,769.00    |              |              |
| Ray tracing dynamic build primitives                                 | 14,988.00 | 15,144.00    | 14,424.00    |              |              |
| Ray tracing active instances                                         | 1,915.00  | 1,915.00     | 1,915.00     |              |              |
| Ray tracing total instances                                          | 1,915.00  | 1,915.00     | 1,915.00     |              |              |
| Mesh draw calls                                                      | 194.95    | 210.00       | 190.00       |              |              |
| Present time                                                         | 0.01 ms   | 0.02 ms      |              |              |              |
| Lights in scene                                                      |           | 88.00        | 88.00        |              |              |
| Scale in scene                                                       |           | 0.00         | 0.00         |              |              |

关闭前 Draw: 12.67ms; Draws: 平均 502; InitViews: 1.28-2.60

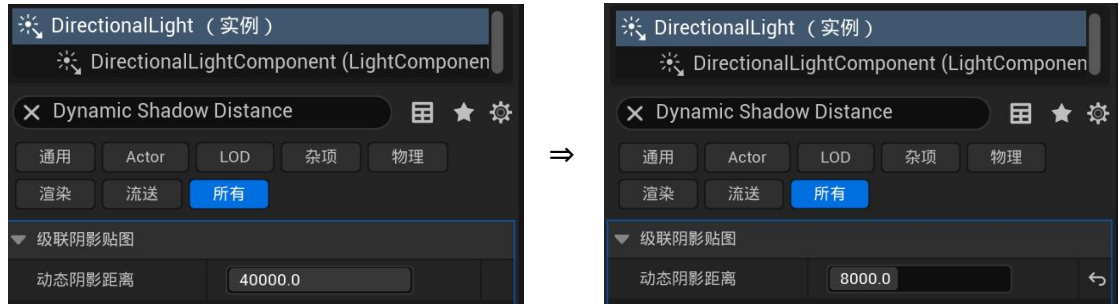
控制台→r.ShadowQuality 0 后:



## 1.1. 灯光层

### Directional Light:

- **Mobility = Stationary**（移动性→固定），允许部分烘焙光照
- **Dynamic Shadow Distance:** 调小（40000 → 8000）



- **Num Dynamic Shadow Cascades:** 4 → 2

## 1.2. 静态网格层

### 1.2.1. 静态小物体关闭阴影

在世界大纲批量选择 Static Mesh / Foliage 等一些较小的不必要制造阴影的物体，取消 Cast Shadow

### 1.2.2. 静态大件用烘焙阴影

目的：静态物阴影烤进“贴图”，运行时不进阴影 Pass

- Edit → Project Settings → Rendering → Lighting, 勾选 **Allow Static Lighting**（允许静态光照）
- Directional Light → Mobility: Stationary（固定），这样做**间接光+静态物阴影**烤进去，近处动态角色还能有实时阴影
- **静态大件本身**，对于地面、外墙、楼板、大建筑件等：
  - Mobility: Static
  - Cast Shadow: true
  - Lighting → Overridden Light Map Res: true（光照贴图分辨率——小建筑件：64、墙/地/楼：256、主角建筑：512）
  - Build Lights Only

构建光照后报警告：

①

 RectLight\_6 服务器性能损耗：由于重叠，无法对静态光源分配阴影贴图，光源将变回动态阴影！

这是由于场景中重叠光源过多，删去不必要的光源或者设为 static 即解决

②

⚠ 未找到重要体积，且场景太大，自动合成的体积无法产生良好的效果。请添加一个紧密包裹的lightmass重要体积来优化场景质量及光照构建时间。

没放体积框，UE 对整个大场景平均烘焙，质量差且极慢

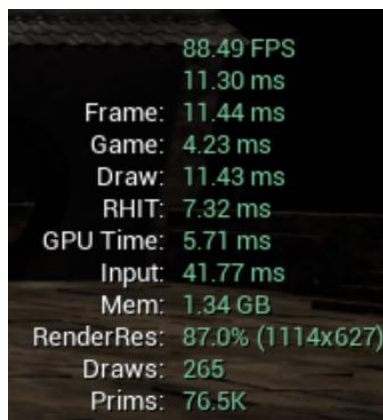
解决方法：

- 放置 Actor → Lightmass Importance Volume
- 拖入场景，缩放包裹住真正需要看清光影的核心区域
- 体积外自动降级，体积内烘焙质量高、速度快。

## 1.3. 阴影优化成果

修改后：

- Draw: 11.43ms
- RHIT: 7.32ms
- Draws: 265



88.49 FPS  
11.30 ms  
Frame: 11.44 ms  
Game: 4.23 ms  
Draw: 11.43 ms  
RHIT: 7.32 ms  
GPU Time: 5.71 ms  
Input: 41.77 ms  
Mem: 1.34 GB  
RenderRes: 87.0% (1114x627)  
Draws: 265  
Prims: 76.5K

- Triangles Drawn: 平均 122K
- Draw calls: 平均 339

| RHI [STATGROUP_RHI]                      |            |              |              |                           |
|------------------------------------------|------------|--------------|--------------|---------------------------|
| Cycle counters (flat)                    |            |              |              |                           |
|                                          | CallCount  | InclusiveAvg | InclusiveMax | ExclusiveAvg ExclusiveMax |
| FRDGPASS Execute                         | 626        | 4.06ms       | 5.01ms       | 3.65ms 4.40ms             |
| Memory Counters                          |            |              |              |                           |
|                                          | UsedMax    | Mem%         | MemPool      | Pool Capacity             |
| Reserved Buffer Memory (Uncommitted)     | 1520.00 MB |              | GPU          |                           |
| Texture 2D Memory                        | 912.21 MB  |              | GPU          |                           |
| Reserved Buffer Memory (Committed)       | 528.00 MB  |              | GPU          |                           |
| Render Target 2D Memory                  | 342.19 MB  |              | GPU          |                           |
| UAV Texture Memory                       | 312.19 MB  |              | GPU          |                           |
| Reserved Texture Memory (Committed)      | 256.00 MB  |              | GPU          |                           |
| RayTracing Acceleration Structure Memory | 67.02 MB   |              | GPU          |                           |
| Vertex Buffer Memory                     | 61.02 MB   |              | GPU          |                           |
| Bindless Resource Heap                   | 41.77 MB   |              | GPU          |                           |
| Structured Buffer Memory                 | 24.72 MB   |              | GPU          |                           |
| Texture Cube Memory                      | 21.31 MB   |              | GPU          |                           |
| Index Buffer Memory                      | 10.27 MB   |              | GPU          |                           |
| Byte Address Buffer Memory               | 4.80 MB    |              | GPU          |                           |
| Texture 3D Memory                        | 2.36 MB    |              | GPU          |                           |
| Bindless Sampler Heap                    | 1.88 MB    |              | GPU          |                           |
| Uniform Buffer Memory                    | 1.07 MB    |              | GPU          |                           |
| Render Target 3D Memory                  | 0.19 MB    |              | GPU          |                           |
| Misc Buffer Memory                       | 0.00 MB    |              | GPU          |                           |
| Draw Indirect Buffer Memory              | 0.00 MB    |              | GPU          |                           |
| Render Target Cube Memory                | 0.00 MB    |              | GPU          |                           |
| Reserved Texture Memory (Uncommitted)    | 0.00 MB    |              | GPU          |                           |
| Counters                                 |            |              |              |                           |
|                                          | Average    | Max          | Min          |                           |
| Resource Descriptors Allocated           |            | 506,567.00   | 506,412.00   |                           |
| Triangles drawn                          | 122,396.08 | 252,858.00   | 78,224.00    |                           |
| Bindless Resource Descriptors Allocated  |            | 6,567.00     | 6,412.00     |                           |
| Sampler Descriptors Allocated            |            | 2,092.00     | 2,092.00     |                           |
| DrawPrimitive calls                      | 339.20     | 353.00       | 331.00       |                           |
| Bindless Sampler Descriptors Allocated   |            | 44.00        | 44.00        |                           |
| Lines drawn                              | 4.00       | 4.00         | 4.00         |                           |

● 前后数据对比

| 指标              | 优化前      | 优化后      | 变化        | 提升幅度                    |
|-----------------|----------|----------|-----------|-------------------------|
| FPS             | 78.95    | 88.49    | ↑ 9.54    | +12.08%                 |
| Draw            | 12.67 ms | 11.43 ms | ↓ 1.24 ms | -9.8%                   |
| Draws           | 502      | 265      | ↓ 237     | -47.2%                  |
| RHIT            | 6.75 ms  | 7.32 ms  | ↑ 0.57 ms | +8.4%（轻微上升，可接受）         |
| Triangles Drawn | 101K     | 122K     | ↑ 21K     | +20.8%（阴影移除后更多几何体进入主渲染） |
| Draw Calls（RHI） | 537      | 339      | ↓ 198     | -36.9%                  |

● 成果解读

- 阴影优化效果显著：Draws 从 502 降至 265（-47.2%），Draw Calls 从 537 降至 339（-36.9%），证明“选择性阴影 + 烘焙 + 删重叠光源”这一系列操作精准命中了场景的核心瓶颈。
- Draw 时间降幅有限：仅从 12.67ms 降到 11.43ms（-9.8%），说明阴影并非渲染线程的唯一负担——剩余 11.43ms 中，静态几何体的碎片化问题（InitViews 中的可见性收集）仍是主要矛盾。
- RHIT 轻微上升：从约 6.75ms 升至 7.32ms，这是因为阴影移除后，原本被阴影 Pass 挡住的几何体进入了主渲染管线，驱动提交压力略有增加，属于正常现象。

2. 静态网格碎片化优化（合批）

2.1. 优化前：

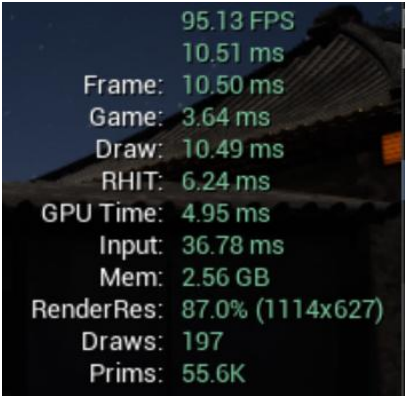
| Cycle counters (flat)         |  | CallCount | InclusiveAvg | InclusiveMax | ExclusiveAvg | ExclusiveMax |
|-------------------------------|--|-----------|--------------|--------------|--------------|--------------|
| Compute View Relevance        |  | 67        | 1.76 ms      | 2.63 ms      | 1.76 ms      | 2.63 ms      |
| View Visibility               |  | 1         | 1.45 ms      | 3.50 ms      | 0.19 ms      | 0.40 ms      |
| Init dynamic shadows          |  | 2         | 0.40 ms      | 0.61 ms      | 0.02 ms      | 0.03 ms      |
| Create WholeScene Shadow      |  | 8         | 0.22 ms      | 0.41 ms      | 0.22 ms      | 0.41 ms      |
| Add View Whole Scene Shadows  |  | 8         | 0.05 ms      | 0.08 ms      | 0.05 ms      | 0.08 ms      |
| GatherShadowPrimitives        |  | 1         | 0.01 ms      | 0.02 ms      | 0.01 ms      | 0.02 ms      |
| Init Projected Shadow         |  | 1         | 0.00 ms      | 0.01 ms      | 0.00 ms      | 0.01 ms      |
| Update Fading                 |  | 3         | 0.00 ms      | 0.01 ms      | 0.00 ms      | 0.01 ms      |
| Update Preshadow Cache        |  | 1         | 0.00 ms      | 0.00 ms      | 0.00 ms      | 0.00 ms      |
| Decompress Occlusion          |  | 1         | 0.00 ms      | 0.00 ms      | 0.00 ms      | 0.00 ms      |
| Counters                      |  | Average   | Max          | Min          |              |              |
| Processed primitives          |  | 2,486.00  | 2,486.00     | 2,486.00     |              |              |
| Visible Raytracing Primitives |  | 1,915.00  | 1,915.00     | 1,915.00     |              |              |
| Occluded primitives           |  | 201.00    | 201.00       | 201.00       |              |              |
| Frustum Culled primitives     |  | 157.00    | 157.00       | 157.00       |              |              |
| Occlusion queries             |  | 51.00     | 51.00        | 51.00        |              |              |
| Visible static mesh elements  |  | 5.00      | 5.00         | 5.00         |              |              |
| Visible dynamic primitives    |  | 4.00      | 4.00         | 4.00         |              |              |

Processed primitives: 2486

## 2.2. 具体操作：

- 按“同网格 + 同材质”筛选：在 World Outliner 中按 Static Mesh 名称排序，筛选出完全相同网格且材质一致的 Actor 组，杜绝不同材质混批。
- 分批合批（Batch）：每组纯净子集选中后，通过 Actor → 合并 Actor → 分批（Batch）生成 ISM/HISM Actor，每批实例数控制在 200 个以下，平衡 Draw Call 与剔除效率。
- 按区域分块：对包围盒太大的同材质同网格组，按视野区块拆分为多个 ISM（如每 100m × 100m 一块），防止大包围盒导致剔除失效。

## 2.3. 合批优化成果：



| Counters                                | Average   | Max        | Min        |
|-----------------------------------------|-----------|------------|------------|
| Resource Descriptors Allocated          |           | 505,944.00 | 505,933.00 |
| Triangles drawn                         | 84,289.98 | 147,926.00 | 60,600.00  |
| Bindless Resource Descriptors Allocated |           | 5,944.00   | 5,933.00   |
| Sampler Descriptors Allocated           |           | 2,092.00   | 2,092.00   |
| DrawPrimitive calls                     | 258.13    | 266.00     | 255.00     |
| Bindless Sampler Descriptors Allocated  |           | 44.00      | 44.00      |
| Lines drawn                             | 4.00      | 4.00       | 4.00       |

| Counters                      | Average | Max    | Min    |
|-------------------------------|---------|--------|--------|
| Processed primitives          | 891.00  | 891.00 | 891.00 |
| Visible Raytracing Primitives | 610.00  | 610.00 | 610.00 |
| Frustum Culled primitives     | 128.00  | 128.00 | 128.00 |
| Occluded primitives           | 99.00   | 99.00  | 99.00  |
| Occlusion queries             | 43.55   | 45.00  | 43.00  |
| Visible static mesh elements  | 11.00   | 11.00  | 11.00  |
| Visible dynamic primitives    | 4.00    | 4.00   | 4.00   |

| 指标                              | 优化前          | 优化后          | 变化幅度     |
|---------------------------------|--------------|--------------|----------|
| Draws（总绘制调用）                    | 256          | 197          | -23.0%   |
| DrawPrimitive Calls             | 339.2        | 258.13       | -23.9%   |
| Processed Primitives<br>（处理图元数） | 2486         | 891          | -64.2%   |
| Draw 耗时                         | 11.43 ms     | 10.49 ms     | -0.94 ms |
| InitViews                       | 1.38-2.24 ms | 1.15-1.83 ms | 稳定下降     |
| FPS                             | 88.49        | 95.13        | +7.5%    |

## 2.4. 踩坑记录：

### ● 第一次分区合批后遇到的主要问题

- **DrawCalls 不降反升：**混合多种 Mesh 与材质合批，被引擎拆成多个组件，状态切换增多，Draws 从约 265 升至 304。
- **耗时与抖动大：**区块包围盒导致“全进全出”剔除失效，HISM 的 BVH 计算增加，渲染线程（Draw 11.29ms）压力大，帧时间抖动更明显。
- **Overdraw 增加：**单批约 100 实例虽在安全线，但混合材质让合批失去意义，无效渲染（Overdraw）增加。

### ● 核心解决方法

- 先从 Saved/Autosaves 恢复备份关卡，重新进行合批操作
- 分类合批：按“同网格 + 同材质”筛选，仅对纯净子集 Batch，实现真正降 DC。
- 同类分块：若同材质网格数量太大或范围（包围盒）太大，按区域拆块，兼顾视锥剔除。

## 3. 其他优化项

鉴于小光源和方向光都是 Stationary 且已烘焙，静态物体的阴影全部在贴图里，理论上运行时零开销。但阴影优化并合批后 再使用 r.ShadowQuality 0 Draw 仍降了 1ms 左右，说明这 1ms 来自动态物体（如角色）在方向光下的实时级联阴影（CSM），以及小光源对动态物的少量实时阴影，可以接受。此时 Draw 耗时还剩余 9ms。

推测剩余耗时来源于 **Lumen 硬件追踪**、**大 ISM 提交与绑定**（合批后单次 DC 数据量大，加光照贴图采样绑定）、**RHIT 线程同步**等。

### 3.1. 关闭硬件光追

Stationary 烘焙已能提供优秀间接光，可以关闭硬件光追。

- Edit → Project Settings → Engine → Rendering → Lumen
- 取消勾选：Use Hardware Ray Tracing when available（在可用时使用硬件光线追踪）



可以看到，关闭硬件光追使用软件光追后，帧率 95.13 → 108.77（+14.3%），RHIT 6.08 ms → 3.91 ms，降 2.17 ms，渲染线程绑定压力缓解，而画质影响极小。

## 四、静态场景性能优化成果总结

以下是将最终数据与最初优化前的基线数据进行量化对比，作为本次针对 Unreal Engine 静态场景的性能优化工作记录的收尾存档。

### 1. 优化前基线数据

初始场景主要瓶颈集中在 CPU 渲染提交侧：Draw 耗时 12.83ms，与 Frame 几乎持平，而 GPU 仅 6.07ms，说明瓶颈不在 GPU 绘制，而在 DrawCall 数量、状态切换与可见性收集。光追相关可见图元高达 1912，静态网格元素仅 6 个，合批严重不足。

### 2. 关键指标量化对比

下表以最初优化前为基线，对比最终稳定状态（帧率 108 FPS 版本）的关键指标。其中【变化】为最终值相对基线的增减量，【提升幅度】以对性能有利的方向计。

| 指标                            | 优化前   | 优化后    | 变化      | 提升幅度   |
|-------------------------------|-------|--------|---------|--------|
| 帧率 (FPS)                      | 79    | 108    | +29     | +36.7% |
| Frame 帧时间 (ms)                | 12.84 | 9.19   | -3.65   | -28.4% |
| Draw 耗时 (ms)                  | 12.83 | 9.33   | -3.50   | -27.3% |
| RHIT (ms)                     | 6.73  | 3.91   | -2.82   | -41.9% |
| GPU Time (ms)                 | 6.07  | 4.84   | -1.23   | -20.3% |
| Draws 总绘制调用                   | 436   | 198    | -238    | -54.6% |
| DrawPrimitive Calls           | 537.2 | 258.13 | -279.07 | -51.9% |
| Triangles Drawn               | 101K  | 55.6K  | -45.4K  | -44.9% |
| Processed Primitives          | 2485  | 891    | -1594   | -64.1% |
| Visible Raytracing Primitives | 1912  | 610    | -1302   | -68.1% |

注：RHIT 降幅最大，得益于剥离硬件光追的实例收集开销；Processed Primitives 与 Raytracing Primitives 大幅减少，得益于静态网格体的规范合批。

### 3. 结论

经过本轮优化，场景帧率由最初的 79 FPS 提升至最终 108 FPS，综合提升约 36.7%；Draw 耗时由 12.83ms 降至 9.33ms，降幅 27.3%。核心收益来自三点：

- 阴影选择性烘焙，消除大量非必要实时阴影 Pass。
- 静态网格按「同网格 + 同材质」规范合批，Processed Primitives 下降 64.1%，可见性收集效率质变。
- 关闭 Lumen 硬件光追改用软件追踪，RHIT 下降 41.9%，渲染线程压力显著缓解。

当前角色站立时帧率稳定在 99~110 FPS 区间，行走时 90+ FPS，物品检视时 70+ FPS（均不影响体验），属引擎渲染管线自然波动。